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Studierichting: Informatica
Oefeningenreeks 2D nr. 16.6

$$
\begin{aligned}
&\lim_{x\to\frac{\pi}{2}} \left(\sin(2x)\tan(3x)\right) = \lim_{x\to\frac{\pi}{2}} \left(\frac{\sin(2x)\sin(3x)}{\cos(3x)}\right) \\
&= \lim_{x\to\frac{\pi}{2}} \left(\frac{\sin(2x)\left(3\sin x - 4\sin^3 x\right)}{4\cos^3 x - 3\cos x}\right) \\
&= \lim_{x\to\frac{\pi}{2}} \left(\frac{2\sin x\cos x\sin x\left(3 - 4\sin^2 x\right)}{\cos x\left(4\cos^2 x - 3\right)}\right) \\
&= \lim_{x\to\frac{\pi}{2}} \left(\frac{2\sin^2 x\left(3 - 4\sin^2 x\right)}{4\cos^2 x - 3}\right) \\
&= \frac{2\left(3-4\right)}{-3} = \frac{2}{3}
\end{aligned}
$$

References